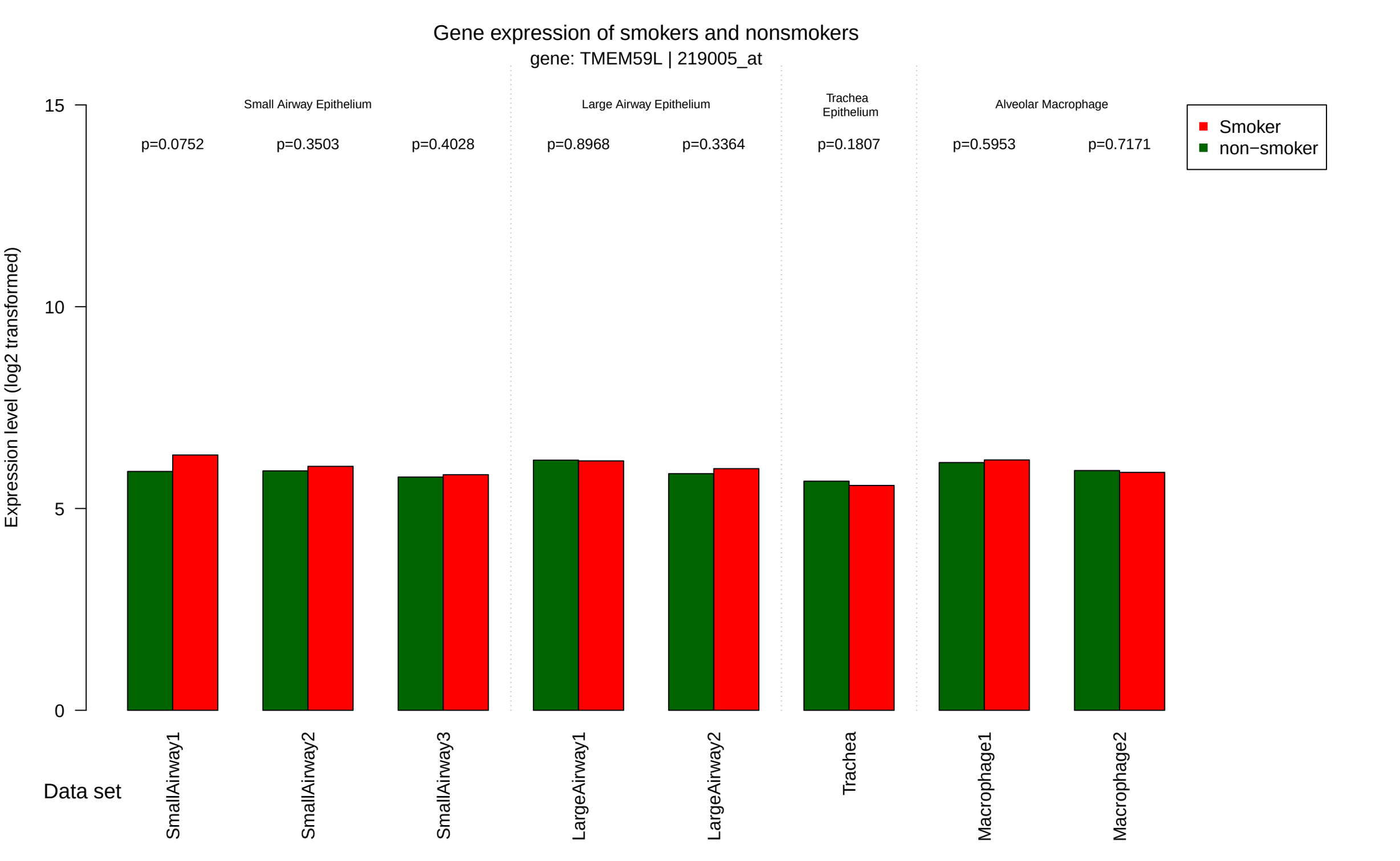
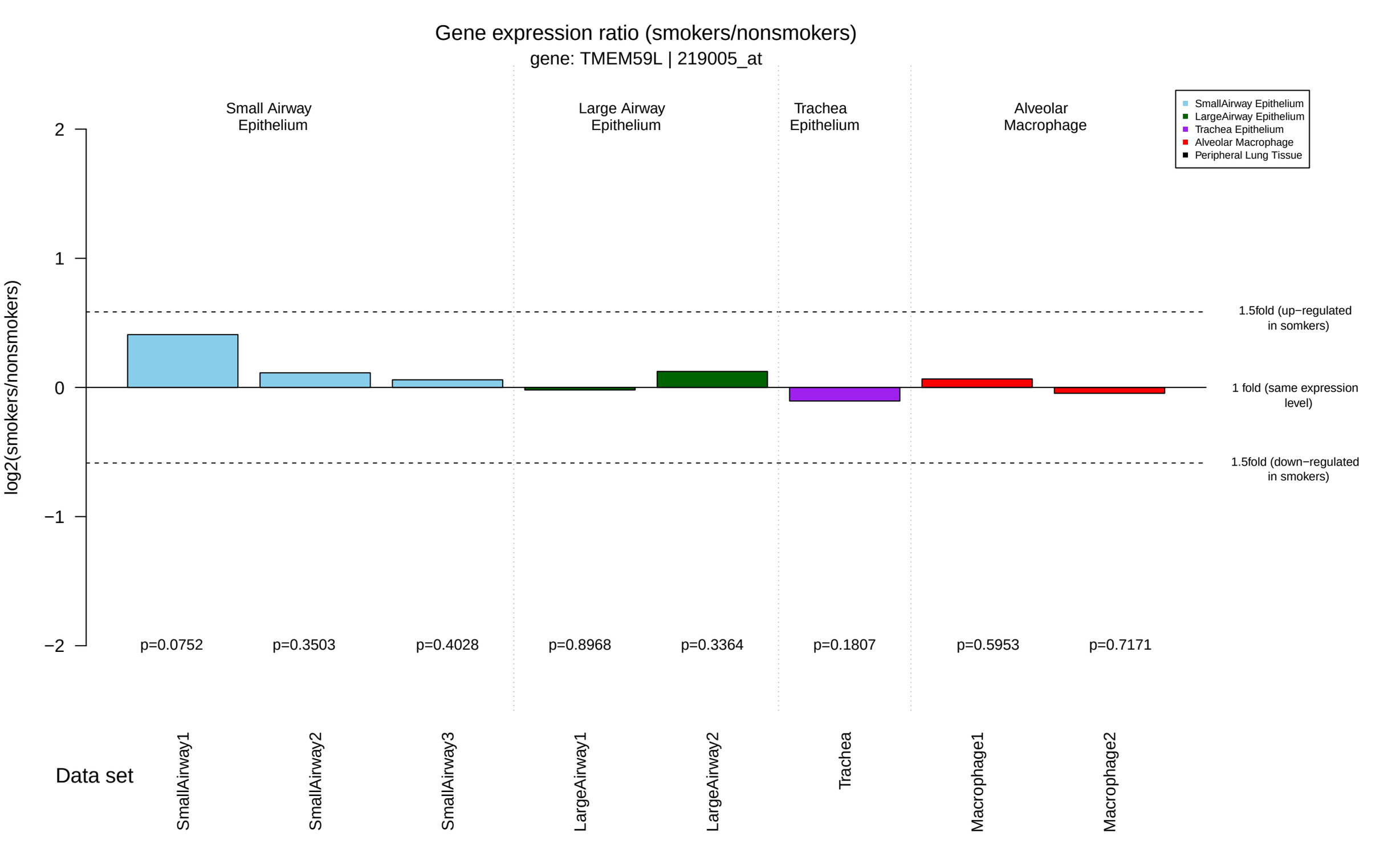
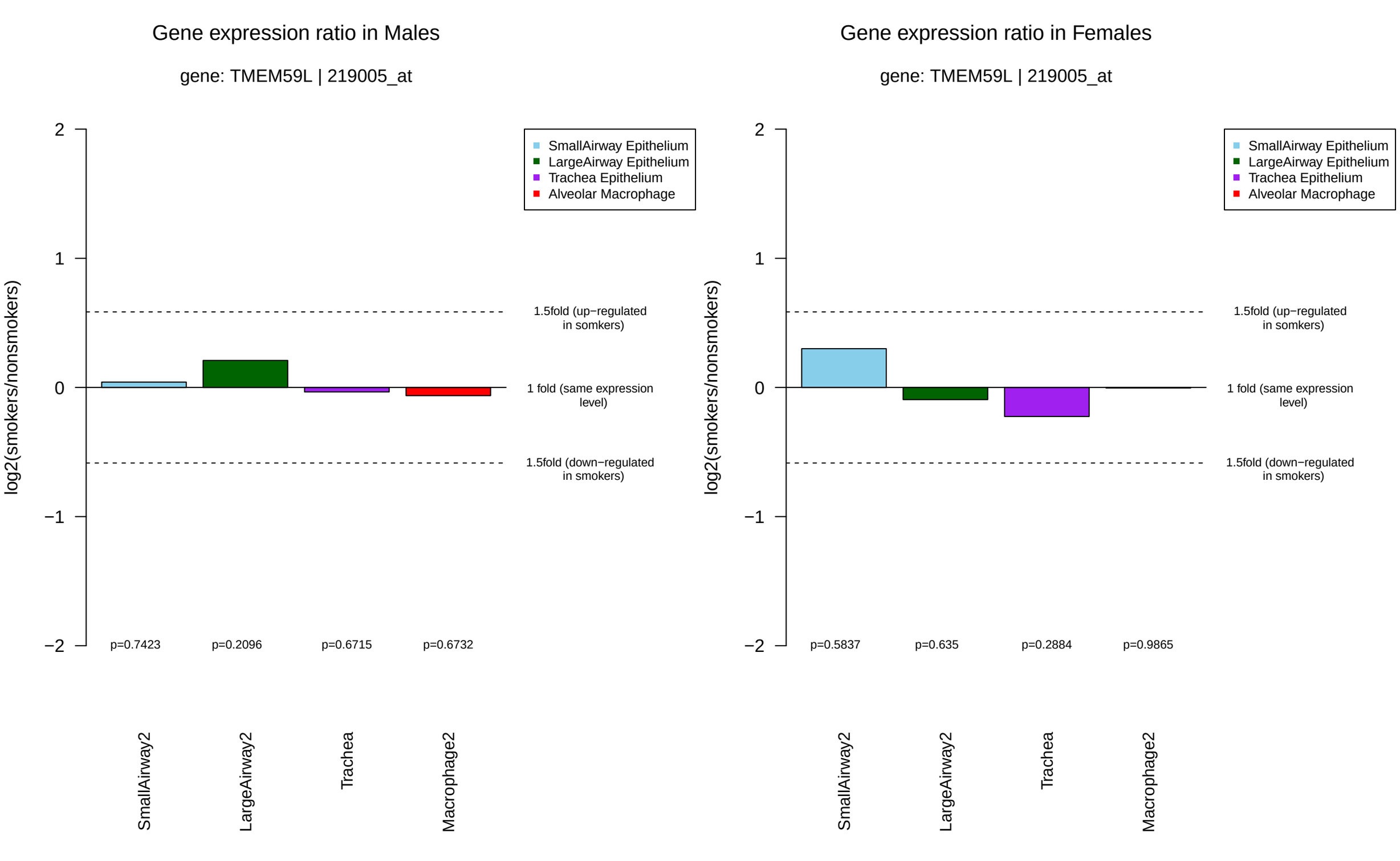
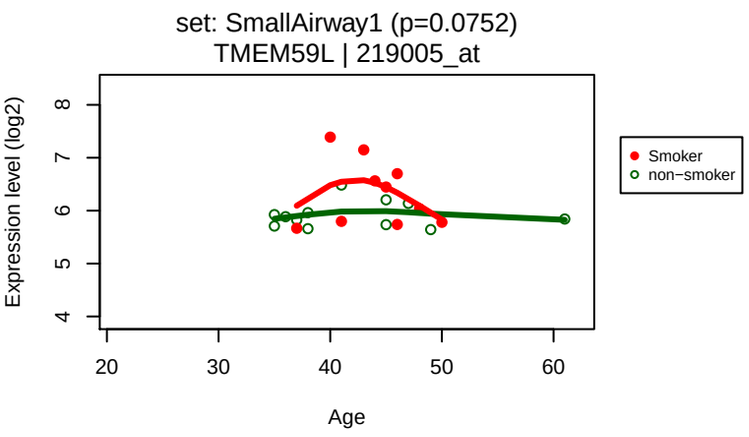
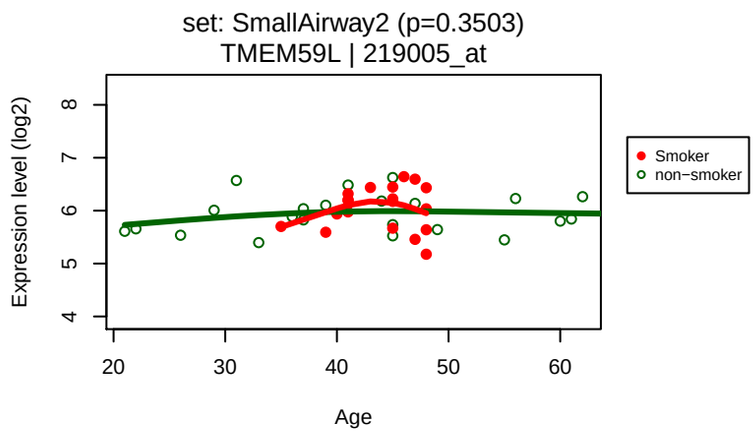
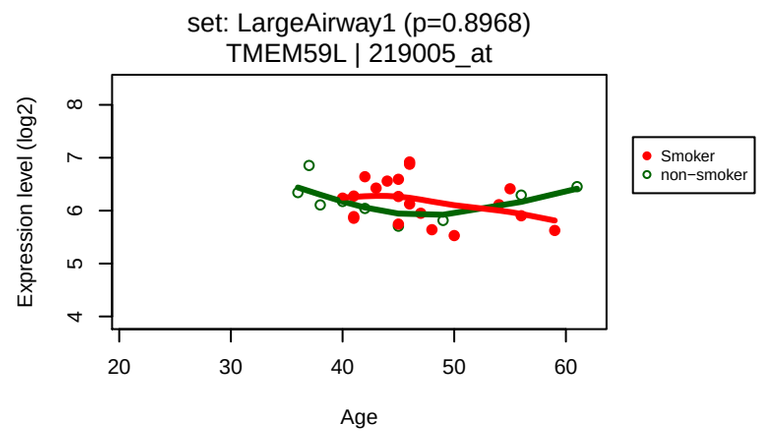

Gene expression of smokers and nonsmokers
gene: TMEM59L | 219005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0752
p=0.3503
p=0.4028
p=0.8968
p=0.3364
p=0.1807
p=0.5953
p=0.7171
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

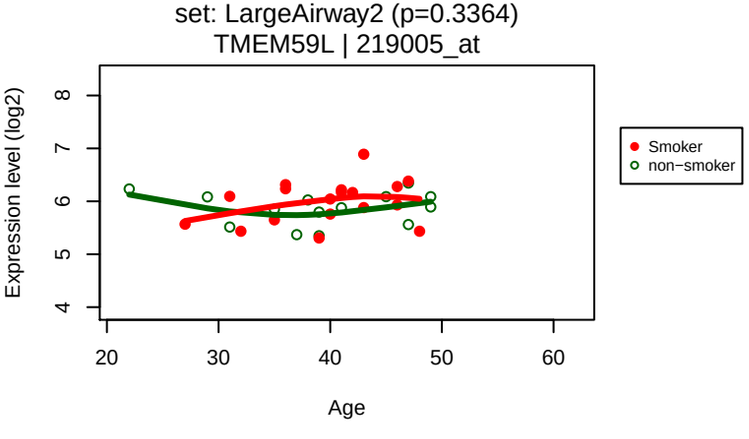

Gene expression ratio (smokers/nonsmokers)
gene: TMEM59L | 219005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0752
p=0.3503
p=0.4028
p=0.8968
p=0.3364
p=0.1807
p=0.5953
p=0.7171
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

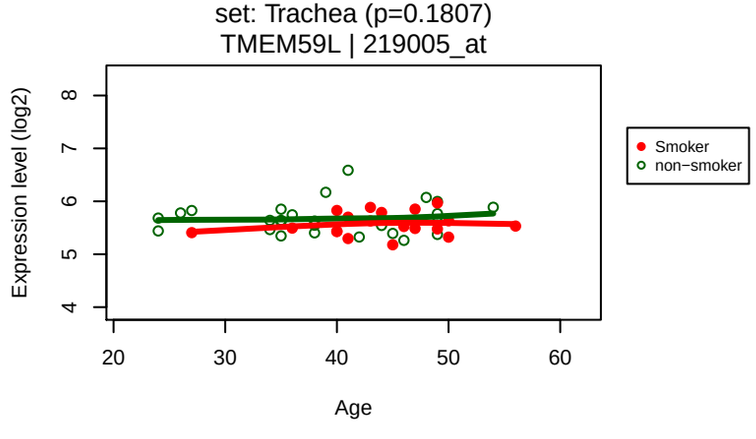

Gene expression ratio in Males
gene: TMEM59L | 219005_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7423
p=0.2096
p=0.6715
p=0.6732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM59L | 219005_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5837
p=0.635
p=0.2884
p=0.9865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

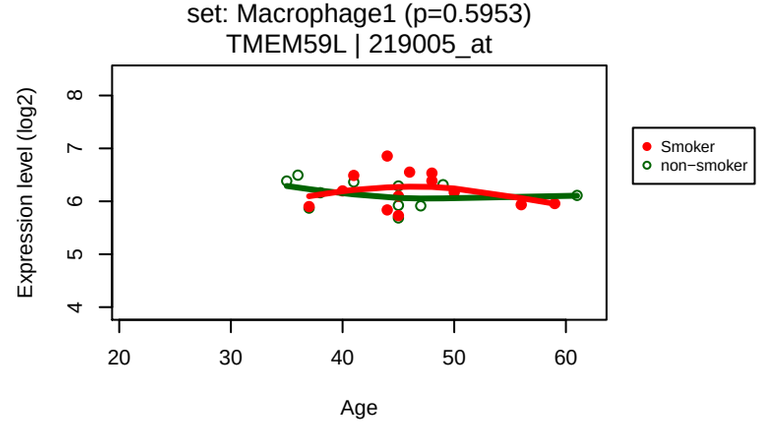

set: SmallAirway1 (p=0.0752)
TMEM59L | 219005_at
Expression level (log2)
Age
Smoker
non−smoker

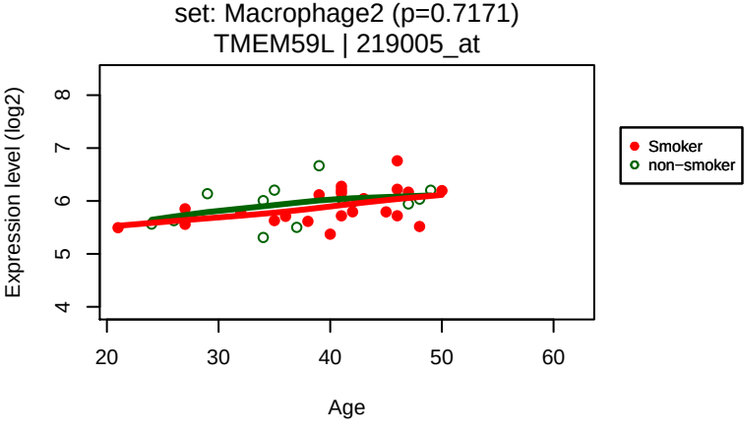

set: SmallAirway2 (p=0.3503)
TMEM59L | 219005_at
Expression level (log2)
Age
Smoker
non−smoker

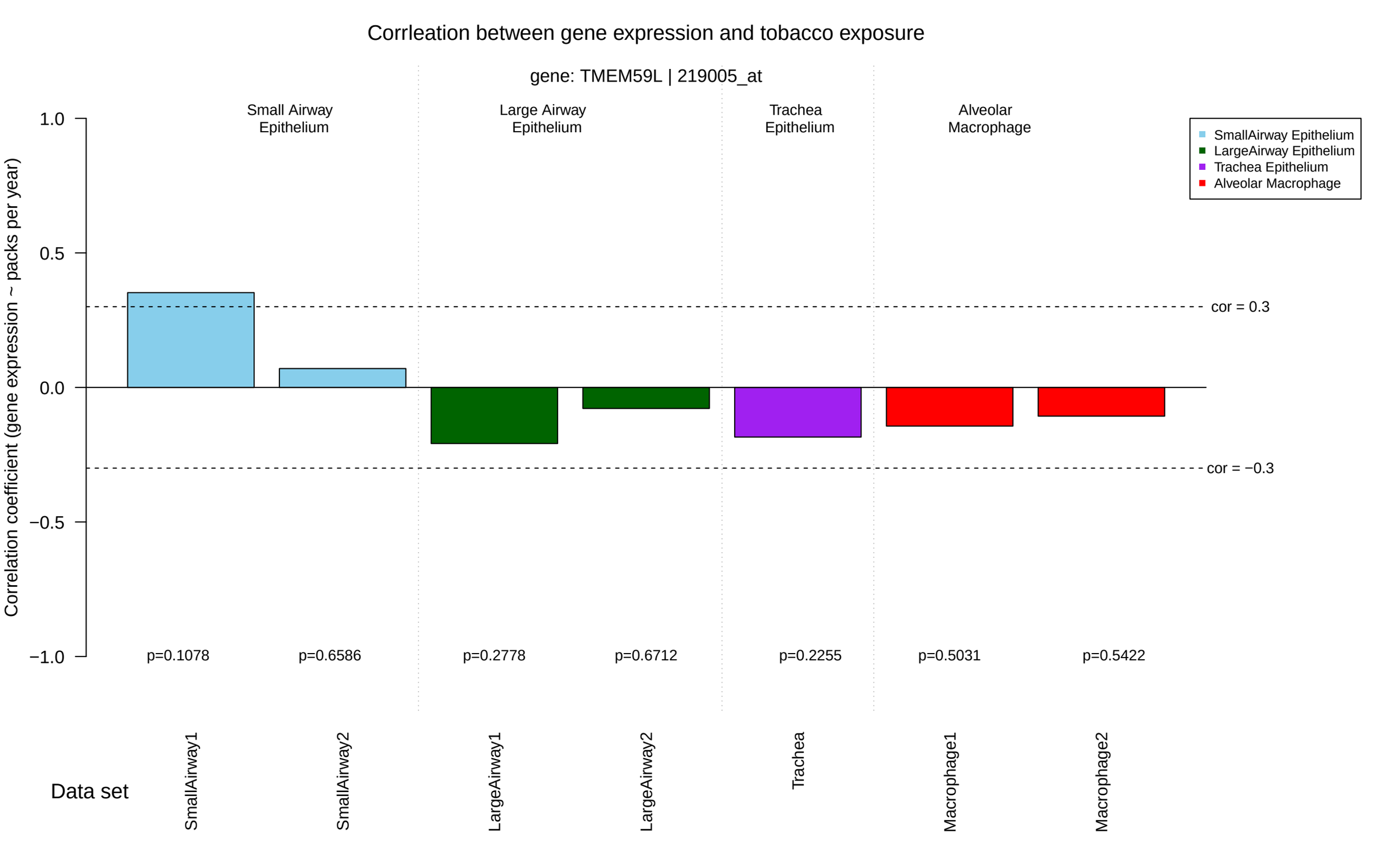

set: LargeAirway1 (p=0.8968)
TMEM59L | 219005_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3364)
TMEM59L | 219005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1807)
TMEM59L | 219005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5953)
TMEM59L | 219005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7171)
TMEM59L | 219005_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM59L | 219005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1078
p=0.6586
p=0.2778
p=0.6712
p=0.2255
p=0.5031
p=0.5422
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2